Nonlinear functions — Part 3

Advanced Math · Math · 86 questions

3.1 f54cf078-dbb9-4605-8b73-28108cb4c7b8 · Hard

The function g is defined by $gx = xx - 2x + 6^2$. The value of $g(7 - w)$ is 0, where w is a constant. What is the sum of all possible values of w?

Student-produced response: _____

3.2 6e108e29-0eee-4507-8a4a-45c4b2c82383 · Hard

A rectangle has an area of 155 square inches. The length of the rectangle is 4 inches less than 7 times the width of the rectangle. What is the width of the rectangle, in inches?

Student-produced response: _____

3.3 397302ed-5a13-4b95-9bee-c626b3a9e221 · Hard

A lab analyst observes a sample of a substance. An exponential model estimates that the mass, in grams, of the sample decreases by 24% every 22.96 minutes. Which of the following equations could represent this model, where M is the estimated mass, in grams, of the sample t minutes after the lab analyst began observing the sample?

A. $M = 1000.24^{t + 22.96}$

B. $M = 1000.24^{\frac{t}{22.96}}$

C. $M = 1000.76^{t + 22.96}$

D. $M = 1000.76^{\frac{t}{22.96}}$

3.4 b0fa9ba5-0ea2-43f1-b456-7834495ae933 · Hard

The functions p and r are defined by the equations shown, where a and b are integer constants such that $a < b$ and $b < 0$. If $y = px$ and $y = rx$ are graphed in the xy-plane, which of the following equations displays, as a constant or coefficient, the y-coordinate of the y-intercept of the graph of the corresponding function?

I. $px = a4.1^{x} + b$
II. $rx = a4.1^{x+b}$

A. I only

B. II only

C. I and II

D. Neither I nor II

3.5 230aed17-3fcc-4941-9290-a69a66f31d95 · Hard

$$y = 2(x - d)(x + d)(x + g)(x - d)$$

In the given equation, d and g are unique positive constants. When the equation is graphed in the xy-plane, how many distinct x-intercepts does the graph have?

A. 4

B. 3

C. 2

D. 1

3.6 019546-DC · Hard

$h(x) = -16x^2 + 100x + 10$

The quadratic function above models the height above the ground h, in feet, of a projectile x seconds after it had been launched vertically. If $y = h(x)$ is graphed in the xy-plane, which of the following represents the real-life meaning of the positive x-intercept of the graph?

A. The initial height of the projectile

B. The maximum height of the projectile

C. The time at which the projectile reaches its maximum height

D. The time at which the projectile hits the ground

3.7 25d9ebe8-a480-4c88-9e5d-9b4803a4780c · Hard

For the exponential function *f*, the value of $f1$ is *k*, where *k* is a constant. Which of the following equivalent forms of the function *f* shows the value of *k* as the coefficient or the base?

A. $fx = 502^{x+1}$

B. $fx = 802^{x}$

C. $fx = 1282^{x-1}$

D. $fx = 2052^{x-2}$

3.8 0f7f7ce8-ffff-4ef3-be02-23e788d84376 · Hard

$$fx = x - 44x - 46$$

The function *f* is defined by the given equation. For what value of *x* does *fx* reach its minimum?

A. 46

B. 45

C. 44

D. -1

3.9 08155-DC · Hard

The population P of a certain city y years after the last census is modeled by the equation below, where r is a constant and P_0 is the population when $y = 0$.

$$P = P_0(1 + r)^y$$

If during this time the population of the city decreases by a fixed percent each year, which of the following must be true?

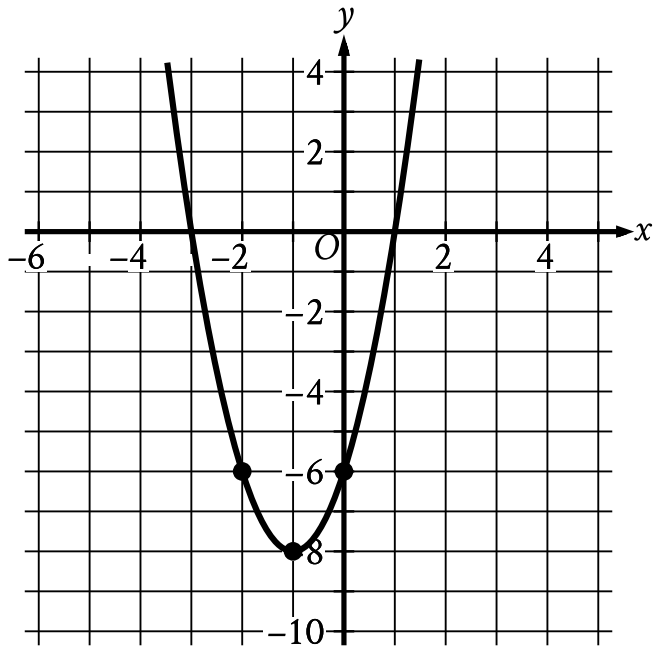
A. $r < -1$

B. $-1 < r < 0$

C. $0 < r < 1$

D. $r > 1$

3.10 a01bd482-116c-496d-b602-9a1cbb2d0dbd · Hard



- The parabola opens upward.
- The vertex is at point (negative 1 comma negative 8).
- The parabola passes through the following points:
  - (negative 2 comma negative 6)
  - (negative 1 comma negative 8)
  - (0 comma negative 6)

The graph of $y = 2x^2 + bx + c$ is shown, where b and c are constants. What is the value of bc ?

Student-produced response: _____

3.11 021312-DC · Hard

$h(x) = x^3 + ax^2 + bx + c$

The function h is defined above, where a, b, and c are integer constants. If the zeros of the function are -5, 6, and 7, what is the value of c ?

Student-produced response: _____

3.12 37d40ac0-74f9-4d4a-8168-1a7d28a874f2 · Hard

$$fx = -6x^2 + 60x - 126$$

The function *f* is defined by the given equation. Which of the following equivalent forms of the equation displays the maximum value of the function as a constant or coefficient?

A. $fx = -6x^2 + 42x + 18x - 126$

B. $fx = -6x(x - 7) + 18(x - 7)$

C. $fx = -6x - 5^2 + 24$

D. $fx = -6x - 7x - 3$

3.13 de664bb9-acef-44d5-af54-80d0da2e25bb · Hard

$$fx = x - 14x + 19$$

The function *f* is defined by the given equation. For what value of *x* does *fx* reach its minimum?

A. -266

B. -19

C. $-\frac{33}{2}$

D. $-\frac{5}{2}$

3.14 b4b30a03-7260-43d7-a5d1-c60b998baa85 · Hard

A park ranger hung squirrel houses each in the shape of a right rectangular prism for fox squirrels. Each house has a height of 11 inches. The length of each house's base is x inches, which is 1 inch more than the width of the house's base. Which function V gives the volume of each house, in cubic inches, in terms of the length of the house's base?

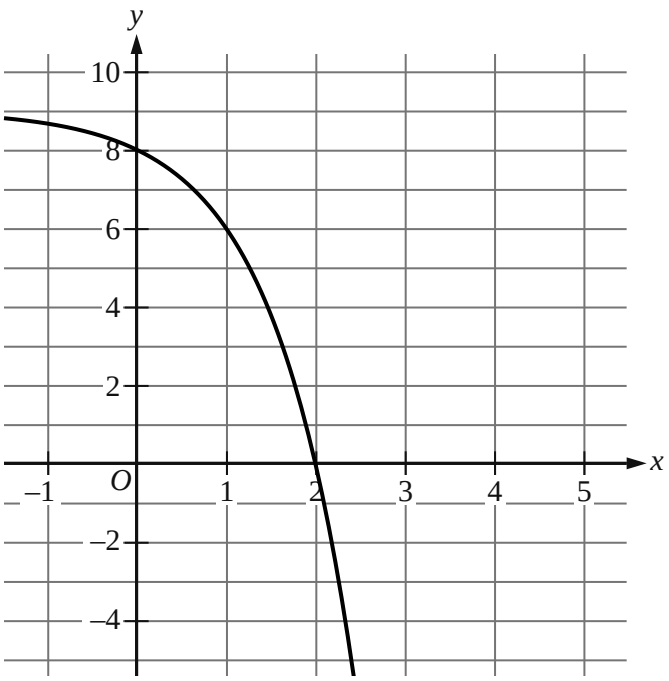
A. $Vx = 11xx - 1$

B. $Vx = 11xx + 1$

C. $Vx = xx + 11x - 1$

D. $Vx = xx + 11x + 1$

3.15 7a673830-c24f-450c-9dbf-0cbce334c119 · Hard



- Moving from left to right:
  - The curve passes from quadrant 2 to quadrant 1 to quadrant 4.
  - In quadrant 2, the curve trends down gradually to the point (0 comma 8).
  - In quadrant 1:
    - The curve trends down gradually to the approximate point (1 comma 6.0).
    - The curve then trends down sharply to the x axis.
  - In quadrant 4, the curve trends down sharply.
- As x decreases, the curve approaches the line y equals 9.
- The curve passes through the following points:
  - (0 comma 8)
  - (1 comma 6)

The graph of $y = fx + 4$ is shown. Which equation defines function f?

A. $fx = -3^x + 1$

B. $fx = -3^x + 5$

C. $fx = -3^x + 8$

D. $f(x) = -3^x + 9$

3.16 917dab83-3019-497d-8e63-5d43c8147242 · Hard

A right rectangular prism has a height of 9 inches. The length of the prism's base is x inches, which is 7 inches more than the width of the prism's base. Which function V gives the volume of the prism, in cubic inches, in terms of the length of the prism's base?

A. $V(x) = xx + 9x + 7$

B. $V(x) = xx + 9x - 7$

C. $V(x) = 9xx + 7$

D. $V(x) = 9xx - 7$

3.17 52a0019a-5b36-4d56-b8de-878dbc9fd5fd · Hard

The function f is defined by $f(x) = a^x + b$, where a and b are constants and $a > 0$. In the xy-plane, the graph of $y = f(x)$ has a y-intercept at $0, -25$ and passes through the point $2, 23$. What is the value of $a + b$?

Student-produced response: _____

3.18 74ac8a77-8102-4458-aafe-4627ed4d7e68 · Hard

The function h is defined by $h(x) = a^x + b$, where a and b are positive constants. The graph of $y = h(x)$ in the xy-plane passes through the points $(0, 10)$ and $(2, 13)$. What is the value of ab?

A. 13

B. 18

C. 20

D. 26

3.19 06375-DC · Hard

A landscaper is designing a rectangular garden. The length of the garden is to be 5 feet longer than the width. If the area of the garden will be 104 square feet, what will be the length, in feet, of the garden?

Student-produced response: _____

3.20 019576-DC · Hard

$M = 1{,}800(1.02)^t$

The equation above models the number of members, M, of a gym t years after the gym opens. Of the following, which equation models the number of members of the gym q quarter years after the gym opens?

A. $M = 1{,}800(1.02)^{\frac{q}{4}}$

B. $M = 1{,}800(1.02)^{4q}$

C. $M = 1{,}800(1.005)^{4q}$

D. $M = 1{,}800(1.082)^{q}$

3.21 cc127122-7a04-46bf-bc9e-396d6ef2a10a · Hard

$$y = 576^{2x+2}$$

The graph of the given equation in the xy-plane has a y-intercept of r, s. Which of the following equivalent equations displays the value of s as a constant, a coefficient, or the base?

A. $y = 331{,}776^{x+1}$

B. $y = 24^{4x+4}$

C. $y = \frac{1}{24}24^{4x+5}$

D. $y = \frac{1}{576}576^{2x+3}$

3.22 63c46f21-221b-43b6-8f48-c9ab2ebf8eb3 · Hard

$$fx = x - 2x + 15$$

The function f is defined by the given equation. For what value of x does fx reach its minimum?

Student-produced response: _____

3.23 2b6d8220-3a75-42df-a439-6f01d10147a0 · Hard

For the function f, $f0 = 86$, and for each increase in x by 1, the value of fx decreases by 80%. What is the value of $f2$?

Student-produced response: _____

3.24 ad9d7217-a0fd-419c-b72d-02d82ef7e7c4 · Hard

A quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object is launched off an elevated surface. The model indicates the object has an initial height of 10 feet above the ground and reaches its maximum height of $1{,}034$ feet above the ground 8 seconds after being launched. Based on the model, what is the height, in feet, of the object above the ground 10 seconds after being launched?

A. 234

B. 778

C. 970

D. $1{,}014$

3.25 038001-DC · Hard

$h(x) = 2(x-4)^2 - 32$

The quadratic function h is defined as shown. In the xy-plane, the graph of $y = h(x)$ intersects the x-axis at the points $(0,0)$ and $(t,0)$, where t is a constant. What is the value of t ?

A. 1

B. 2

C. 4

D. 8

3.26 17ec22bd-b74b-4395-bda1-506081819636 · Hard

In the *xy*-plane, a parabola has vertex $(9, -14)$ and intersects the *x*-axis at two points. If the equation of the parabola is written in the form $y = ax^2 + bx + c$, where *a*, *b*, and *c* are constants, which of the following could be the value of $a + b + c$?

A. -23

B. -19

C. -14

D. -12

3.27 e03b8f0c-00cb-46f3-af40-2ba9b200ab9c · Hard

The function *f* is defined by $fx = ax^2 + bx + c$, where *a*, *b*, and *c* are constants. The graph of $y = fx$ in the *xy*-plane passes through the points $7, 0$ and $-3, 0$. If *a* is an integer greater than 1 , which of the following could be the value of $a + b$?

A. -6

B. -3

C. 4

D. 5

3.28 025785-DC · Hard?xml version="1.0" encoding="UTF-8"?

x	p(x)
−2	5
−1	0
0	−3
1	−1
2	0

The table above gives selected values of a polynomial function p. Based on the values in the table, which of the following must be a factor of p ?

A. $(x-3)$

B. $(x+3)$

C. $(x-1)(x+2)$

D. $(x+1)(x-2)$

3.29 65f2a8b0-dac1-4676-af53-3ed1d29659c3 · Hard

The function g is defined by $gx = \frac{x}{a} - 14$, where $a < 0$. What is the product of $g15a$ and $g7a$?

Student-produced response: _____

3.30 08160-DC · Hard

Growth of a Culture of Bacteria

Day	Number of bacteria per milliliter at end of day
1	2.5×10^5
2	5.0×10^5
3	1.0×10^6

A culture of bacteria is growing at an exponential rate, as shown in the table above. At this rate, on which day would the number of bacteria per milliliter reach 5.12×10^8?

A. Day 5

B. Day 9

C. Day 11

D. Day 12

3.31 5291f5d1-e675-4632-9051-76a6be583d50 · Hard

Which of the following functions has(have) a minimum value at -3?

I. $f(x) = -6(3)^x - 3$

II. $g(x) = -3(6)^x$

A. I only

B. II only

C. I and II

D. Neither I nor II

3.32 e4d534f2-3a87-4887-9602-3feff6694dcb · Hard

A quadratic function models a projectile's height, in meters, above the ground in terms of the time, in seconds, after it was launched. The model estimates that the projectile was launched from an initial height of 7 meters above the ground and reached a maximum height of 51.1 meters above the ground 3 seconds after the launch. How many seconds after the launch does the model estimate that the projectile will return to a height of 7 meters?

A. 3

B. 6

C. 7

D. 9

3.33 7224c605-0224-443d-8b93-f567251f7a19 · Hard

The product of two positive integers is 462. If the first integer is 5 greater than twice the second integer, what is the smaller of the two integers?

Student-produced response: _____

3.34 22bb3d70-ebff-46aa-8f1d-07a231d876e8 · Hard

$$f(x) = x^2 - 48x + 2{,}304$$

What is the minimum value of the given function?

Student-produced response: _____

3.35 e5403cac-ceb7-4643-b03d-c3f63fe73713 · Hard

$$f(x) = 271.20^{\frac{x}{3}}$$

For the given function f, the value of $f(x)$ increases by $p\%$ for every increase of x by 6. What is the value of p?

Student-produced response: _____

3.36 05720-DC · Hard

The surface area of a cube is $6\left(\frac{a}{4}\right)^2$, where a is a positive constant. Which of the following gives the perimeter of one face of the cube?

A. $\frac{a}{4}$

B. a

C. $4a$

D. $6a$

3.37 f78208f1-bb88-4092-9147-edda34ecda72 · Hard

Function f is a quadratic function where $f(-20) = 0$ and $f(-4) = 0$. The graph of $y = f(x)$ in the xy-plane has a vertex at $(r, -64)$. What is the value of r?

Student-produced response: _____

3.38 ace58eff-0b88-4468-aeb9-2f5ff6974cd4 · Hard

The function f is defined by $f(x) = a(2.2^x + 2.2^b)$, where a and b are integer constants and $0 < a < b$. The functions g and h are equivalent to function f, where k and m are constants. Which of the following equations displays the y-coordinate of the y-intercept of the graph of $y = f(x)$ in the xy-plane as a constant or coefficient?

I. $g(x) = a(2.2^x) + k$
II. $h(x) = a(2.2^x) + m$

A. I only

B. II only

C. I and II

D. Neither I nor II

3.39 c52ea1c2-4bf1-4d4c-b86b-2dc95f66577b · Hard

The function g is defined by $gx = x + 14t - x$, where t is a constant. In the xy-plane, the graph of $y = gx$ passes through the point $24, 0$. What is the value of $g0$?

Student-produced response: _____

3.40 2ae9a93c-e2a5-4e1e-9012-cdaf6592e09a · Hard

$$fx = 4x^2 + 64x + 262$$

The function g is defined by $gx = fx + 5$. For what value of x does gx reach its minimum?

A. -13

B. -8

C. -5

D. -3

3.41 df267826-c54f-44df-bbc4-03bc6a5415ad · Hard

At the time that an article was first featured on the home page of a news website, there were 40 comments on the article. An exponential model estimates that at the end of each hour after the article was first featured on the home page, the number of comments on the article had increased by 190% of the number of comments on the article at the end of the previous hour. Which of the following equations best represents this model, where C is the estimated number of comments on the article t hours after the article was first featured on the home page and $t \leq 4$?

A. $C = 401.19^t$

B. $C = 401.9^t$

C. $C = 4019^t$

D. $C = 402.9^t$

3.42 b47b1926-6e95-4552-8fc8-d6ec1af4c692 · Hard

x	y
21	-8
23	8
25	-8

The table shows three values of x and their corresponding values of y, where $y = fx + 4$ and f is a quadratic function. What is the y-coordinate of the y-intercept of the graph of $y = fx$ in the xy-plane?

Student-produced response: _____

3.43 68f5f6ee-c704-4500-882b-ba9cb2d1a598 · Hard

The functions f and g are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the maximum value of the function it defines, where $x \geq 0$?

I. $fx = 330.4^{x+3}$
II. $gx = 330.160.4^{x-2}$

A. I only

B. II only

C. I and II

D. Neither I nor II

3.44 2812bef6-4e2d-4eae-b382-81985e9c9a12 · Hard

$$Pt = 2601.04^{\frac{6}{4}t}$$

The function P models the population, in thousands, of a certain city t years after 2003. According to the model, the population is predicted to increase by 4% every n months. What is the value of n?

A. 8

B. 12

C. 18

D. 72

3.45 bf8e1345-911b-42f4-ad9c-688a5ce90cb6 · Hard

A machine launches a softball from ground level. The softball reaches a maximum height of 51.84 meters above the ground at 1.8 seconds and hits the ground at 3.6 seconds. Which equation represents the height above ground h, in meters, of the softball t seconds after it is launched?

A. $h = -t^2 + 3.6$

B. $h = -t^2 + 51.84$

C. $h = -16t - 1.8^2 - 3.6$

D. $h = -16t - 1.8^2 + 51.84$

3.46 bb4ceae0-91ef-4a7e-8a48-9eecde3713f6 · Hard

The function f gives the product of a number, x, and a number that is 91 more than x. Which equation defines f?

A. $fx = x^2 + x + 91$

B. $fx = x^2 + 91$

C. $fx = x^2 + 91x$

D. $fx = x^2 + 91x + 91$

3.47 8defe594-1c91-42f1-b36d-c8fc7e515441 · Hard

The function f is defined by $f(x) = (-8)(2)^x + 22$. What is the y-intercept of the graph of $y = f(x)$ in the xy-plane?

A. $(0, 14)$

B. $(0, 2)$

C. $(0, 22)$

D. $(0, -8)$

3.48 029595-DC · Hard

$$f(x) = -500x^2 + 25{,}000x$$

The revenue $f(x)$, in dollars, that a company receives from sales of a product is given by the function f above, where x is the unit price, in dollars, of the product. The graph of $y = f(x)$ in the xy-plane intersects the x-axis at 0 and a. What does a represent?

A. The revenue, in dollars, when the unit price of the product is $0

B. The unit price, in dollars, of the product that will result in maximum revenue

C. The unit price, in dollars, of the product that will result in a revenue of $0

D. The maximum revenue, in dollars, that the company can make

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3.49 5b4efd5b-3a24-4e83-93ed-1065e3a90515 · Hard

According to a model, the estimated volume of a certain tree in a forest increases by 13.9% for every increase of 100 centimeters in the tree's height. The equation $y = ab^{\frac{x}{100}}$ represents this model, where y represents the estimated volume, in cubic centimeters, of this tree in the forest, x represents the tree's height, in centimeters, and a and b are constants. In this equation, what is the value of b?

A. 0.139

B. 1

C. 1.139

D. 13.9

3.50 029650-DC · Hard

x	$f(x)$
1	a
2	a^5
3	a^9

For the exponential function f, the table above shows several values of x and their corresponding values of $f(x)$, where a is a constant greater than 1. If k is a constant and $f(k) = a^{29}$, what is the value of k ?

Student-produced response: _____

3.51 08364-DC · Hard

The function f is defined by $f(x)=(x+3)(x+1)$. The graph of f in the xy-plane is a parabola. Which of the following intervals contains the x-coordinate of the vertex of the graph of f ?

A. $-4<x<-3$

B. $-3<x<1$

C. $1<x<3$

D. $3<x<4$

3.52 c061fa3e-bd9b-463c-94fa-bfb0819fdf17 · Hard

$$fx = 1.84^{\frac{x}{4}}$$

The function f is defined by the given equation. The equation can be rewritten as $fx = 1 + \frac{p}{100}^{x}$, where p is a constant. Which of the following is closest to the value of p?

A. 16

B. 21

C. 46

D. 96

3.53 158b94a1-9ef6-432b-9a5c-1435e1662fe7 · Hard

x	gx
-27	3
-9	0
21	5

The table shows three values of x and their corresponding values of gx, where $gx = \frac{fx}{x+3}$ and f is a linear function. What is the y-intercept of the graph of $y = fx$ in the xy-plane?

A. $0, 36$

B. $0, 12$

C. $0, 4$

D. $0, -9$

3.54 27964a59-cacc-4c94-b077-538b9a79dbb5 · Hard

$$f(x) = (x - 1)(x + 3)(x - 2)$$

In the xy-plane, when the graph of the function f, where $y = f(x)$, is shifted up 6 units, the resulting graph is defined by the function g. If the graph of $y = g(x)$ crosses through the point $4, b$, where b is a constant, what is the value of b?

Student-produced response: _____

3.55 cce2d985-f0ec-4127-81ef-a47c95e786ed · Hard

The function f is a quadratic function. In the xy-plane, the graph of $y = fx$ has a vertex at $1, 9$ and passes through the points $2, 33$ and $-1, 105$. What is the value of $f(-2) - f(0)$?

A. 81

B. 129

C. 192

D. 225

3.56 0fbeb13e-6775-487f-8f08-a6bf51a5bad6 · Hard

Two variables, x and y, are related such that for each increase of 1 in the value of x, the value of y increases by a factor of 4. When $x = 0$, $y = 200$. Which equation represents this relationship?

A. $y = 4x^{200}$

B. $y = 4200^{x}$

C. $y = 200x^{4}$

D. $y = 2004^{x}$

3.57 e34a2287-9f25-422a-b964-03bc949e8c94 · Hard

Function f is defined by $fx = -a^{x} + b$, where a and b are constants. In the xy-plane, the graph of $y = fx - 15$ has a y-intercept at $0, -\frac{99}{7}$. The product of a and b is $\frac{65}{7}$. What is the value of a?

Student-produced response: _____

3.58 020462-DC · Hard

What is the minimum value of the function f defined by $f(x) = (x-2)^2 - 4$?

A. -4

B. -2

C. 2

D. 4

3.59 217d73a6-2b65-4411-bdbd-71f2d68adf48 · Hard

$$fx = 9{,}0000.66^{x}$$

The given function f models the number of advertisements a company sent to its clients each year, where x represents the number of years since 1997, and $0 \leq x \leq 5$. If $y = fx$ is graphed in the xy-plane, which of the following is the best interpretation of the y-intercept of the graph in this context?

A. The minimum estimated number of advertisements the company sent to its clients during the 5 years was $1{,}708$.

B. The minimum estimated number of advertisements the company sent to its clients during the 5 years was $9{,}000$.

C. The estimated number of advertisements the company sent to its clients in 1997 was $1{,}708$.

D. The estimated number of advertisements the company sent to its clients in 1997 was $9{,}000$.

3.60 5f865b4c-8ea2-42c5-aae5-8d455a024fd0 · Hard

The functions g and h are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?

I. $gx = 181.161.4^{x+2}$
II. $hx = 181.4^{x+4}$

A. I only

B. II only

C. I and II

D. Neither I nor II

3.61 821bb037-c906-433b-a30b-6197a4337fb6 · Hard

$$fx = 5{,}4700.64^{\frac{x}{12}}$$

The function *f* gives the value, in dollars, of a certain piece of equipment after *x* months of use. If the value of the equipment decreases each year by *p*% of its value the preceding year, what is the value of *p*?

A. 4

B. 5

C. 36

D. 64

3.62 5d6ca18c-fd03-4a2f-89ab-d392b028d18b · Hard

Function *f* is defined by $fx = -a^x + b$, where *a* and *b* are constants. In the *xy*-plane, the graph of $y = fx - 12$ has a *y*-intercept at $0, -\frac{75}{7}$. The product of *a* and *b* is $\frac{320}{7}$. What is the value of *a*?

Student-produced response: _____

3.63 019540-DC · Hard

The population of a town is currently 50,000, and the population is estimated to increase each year by 3% from the previous year. Which of the following equations can be used to estimate the number of years, t, it will take for the population of the town to reach 60,000 ?

A. $50{,}000 = 60{,}000(0.03)^t$

B. $50{,}000 = 60{,}000(3)^t$

C. $60{,}000 = 50{,}000(0.03)^t$

D. $60{,}000 = 50{,}000(1.03)^t$

3.64 845ca085-d722-4829-90e4-3d2962743b6a · Hard

$$gx = \frac{x^2 - x - a}{x^3 - x - b}$$

The function g is defined by the given equation, where a and b are constants. In the xy-plane, the graph of $y = gx$ passes through the point 022, and $g-22 = 0$. What is the value of b ?

A. 23

B. 22

C. -22

D. -23

3.65 a5817e09-28f5-4ad8-9f99-d3d5b3651286 · Hard

When the quadratic function f is graphed in the xy-plane, where $y = fx$, its vertex is $-3, 6$. One of the x-intercepts of this graph is $-\frac{17}{4}, 0$. What is the other x-intercept of the graph?

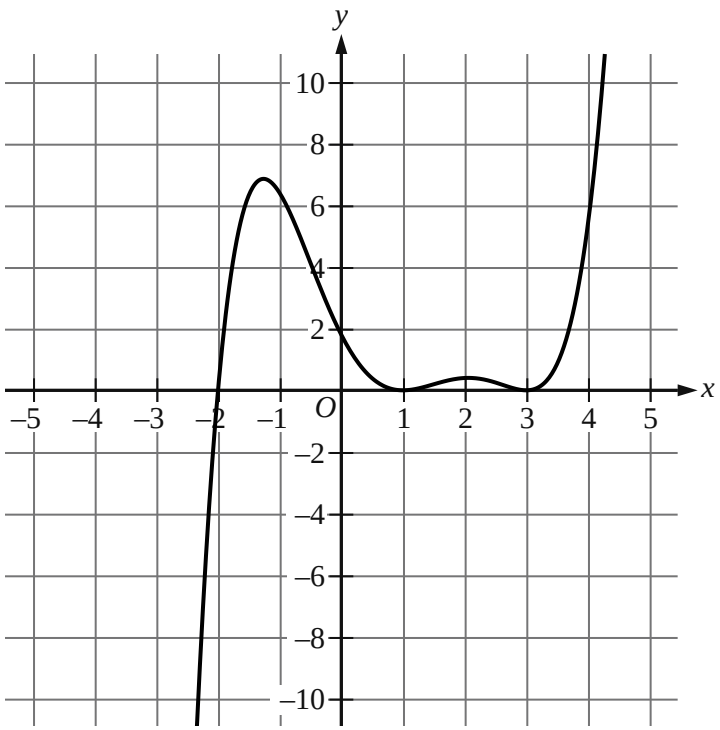
A. $-\frac{29}{4}, 0$

B. $-\frac{7}{4}, 0$

C. $\frac{5}{4}, 0$

D. $\frac{17}{4}, 0$

3.66 f4d42535-f2e1-455c-b197-383e00a2a878 · Hard



- Moving from left to right, the curve passes from quadrant 3 to quadrant 2 to quadrant 1.
- The curve has 2 relative maximums and 2 relative minimums.
- The curve crosses the x axis at the point (negative 2 comma 0).
- The curve touches the x axis at the following points:
  - (1 comma 0)
  - (3 comma 0)

The graph of $y = gx$ is shown. If $gx + 28 = \frac{1}{10}x - a^2 x - b^2 x - c$, where a, b, and c are constants, which of the following could be the value of b?

A. 31

B. 3

C. -2

D. -25

3.67 4fbe07d0-3558-4621-b4b3-5d8c574429ee · Hard

$$ht = -16t^2 + b$$

The function h estimates a ball's height, in feet, above a floor t seconds after the ball is released, where b is a constant. The function estimates that the ball is 19.36 feet above the floor when it is released at $t = 0$. How many seconds after being released does the function estimate the ball will reach the floor?

Student-produced response: _____

3.68 e5580e8a-65b2-4cef-9984-856a4c58c355 · Hard

Function f is defined by $fx = x + 6x + 5x + 1$. Function g is defined by $gx = fx - 1$. The graph of $y = gx$ in the xy-plane has x-intercepts at $a, 0$, $b, 0$, and $c, 0$, where a, b, and c are distinct constants. What is the value of $a + b + c$?

A. -15

B. -9

C. 11

D. 15

3.69 c9709019-d048-4409-a506-d771fa061265 · Hard

The functions f and g are defined by the given equations.

$$fx = 3 + |-2x - x^2|$$

$$gw = \left|\frac{-w}{w-1}\right| - w + 5$$

If $f(-4) = c$, where c is a constant, what is the value of gc?

Student-produced response: _____

3.70 7bb5425b-4812-479b-b73a-acf842444820 · Hard

A model estimates that at the end of each year from 2015 to 2020, the number of squirrels in a population was 150% more than the number of squirrels in the population at the end of the previous year. The model estimates that at the end of 2016, there were 180 squirrels in the population. Which of the following equations represents this model, where n is the estimated number of squirrels in the population t years after the end of 2015 and $t \leq 5$?

A. $n = 721.5^t$

B. $n = 722.5^t$

C. $n = 1801.5^t$

D. $n = 1802.5^t$

3.71 10bfa82a-f80c-43fa-bf29-485d8932309c · Hard

The area of a rectangular banner is $2{,}661$ square inches. The banner's length x, in inches, is 24 inches longer than its width, in inches. Which equation represents this situation?

A. $0 = x^2 - 24x - 2{,}661$

B. $0 = x^2 - 24x + 2{,}661$

C. $0 = x^2 + 24x - 2{,}661$

D. $0 = x^2 + 24x + 2{,}661$

3.72 c33f8edf-4790-4382-963c-36f09a033f73 · Hard

The function f is defined by $fx = a\sqrt{x + b}$, where a and b are constants. In the xy-plane, the graph of $y = fx$ passes through the point $-24, 0$, and $f24 < 0$. Which of the following must be true?

A. $f0 = 24$

B. $f0 = -24$

C. $a > b$

D. $a < b$

3.73 212a6362-e804-4f52-add2-1e4b1d20094c · Hard

The function f is defined by $fx = x - 6x - 2x + 6$. In the xy-plane, the graph of $y = gx$ is the result of translating the graph of $y = fx$ up 4 units. What is the value of $g0$?

Student-produced response: _____

3.74 581cd95e-37cf-4835-9e46-c492ec9a8f24 · Hard

$$ft = 55t - 2t^2$$

The function f is defined by the given equation. The function g is defined by $gt = ft + 3$. Which expression represents the maximum value of gt?

A. $3 + \frac{55}{2}^2$

B. $3 + 2\frac{55}{4}^2$

C. $3 - 2\frac{55}{4}^2$

D. $3 - \frac{55}{2}^2$

3.75 406ed77a-b110-44a7-9d44-1c6ff1aa5918 · Hard

$$fx = 2722^x$$

The function f is defined by the given equation. If $hx = fx - 4$, which of the following equations defines function h?

A. $hx = 172^x$

B. $hx = 682^x$

C. $hx = 27216^x$

D. $hx = 2728^x$

3.76 29361323-e4f7-4672-a65d-d394fcdb43c0 · Hard

A certain bowl was sold online for prices varying between $\$ 7$ and $\$ 13$. The number of bowls sold can be modeled by a quadratic function based on the price, where 5 bowls were sold for $\$ 12$ each and the maximum of 9 bowls were sold for $\$ 10$ each. According to the model, what is the number of bowls that were sold for $\$ 8$ each?

Student-produced response: _____

3.77 1456688b-dc2f-4218-865c-e9bd32de6518 · Hard

The function h is defined by $h(x) = a^x + b$, where a and b are positive constants. The graph of $y = h(x)$ in the xy-plane passes through the points $(0, 10)$ and $(-2, \frac{325}{36})$. What is the value of ab?

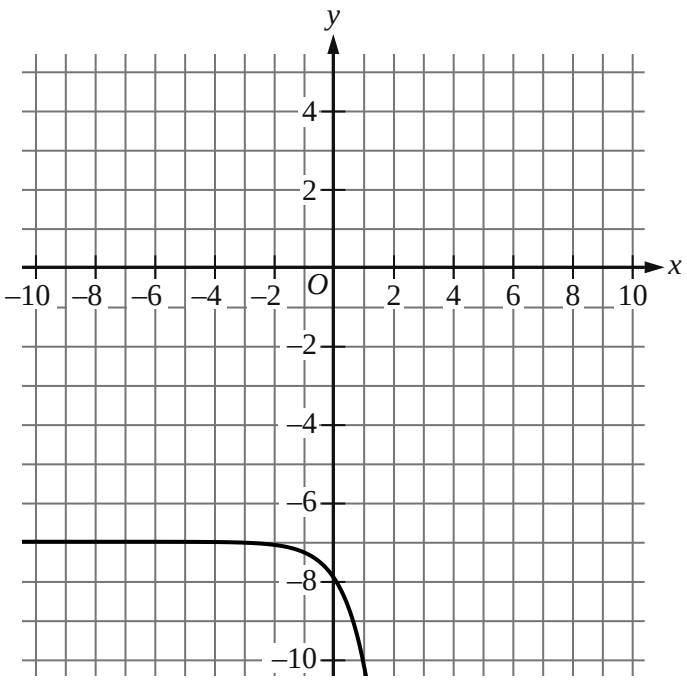
A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. 54

D. 60

3.78 92038ced-e8e7-4472-a025-2fd98e61dd44 · Hard



- Moving from left to right, the curve passes from quadrant 3 to quadrant 4.
- As x decreases, the curve approaches line y equals negative 7.
- The curve passes through the following approximate points:
  - (negative 1 comma negative 7.3)
  - (0 comma negative 7.9)

The graph of $y = fx$ is shown, where $fx = ab^x + c$, and a, b, and c are constants. For how many values of x does $fx = 0$?

A. Three

B. Two

C. One

D. Zero

3.79 f8a81d21-4cde-4a6a-bab6-e67dcb97b10f · Hard

An auditorium has seats for $1{,}800$ people. Tickets to attend a show at the auditorium currently cost $\$ 4.00$. For each $\$ 1.00$ increase to the ticket price, 100 fewer tickets will be sold. This situation can be modeled by the equation $y = -100x^2 + 1{,}400x + 7{,}200$, where x represents the increase in ticket price, in dollars, and y represents the revenue, in dollars, from ticket sales. If this equation is graphed in the xy-plane, at what value of x is the maximum of the graph?

A. 4

B. 7

C. 14

D. 18

3.80 b2aba242-20bd-454b-aaca-55c42bdd0501 · Hard

$$y = x^2 - 14x + 22$$

The given equation relates the variables x and y. For what value of x does the value of y reach its minimum?

Student-produced response: _____

3.81 c99a75dd-0256-4d13-b211-eb05d63ef00d · Hard

Square P has a side length of x inches. Square Q has a perimeter that is 176 inches greater than the perimeter of square P. The function f gives the area of square Q, in square inches. Which of the following defines f?

A. $fx = x + 44^2$

B. $fx = x + 176^2$

C. $fx = 176x + 44^2$

D. $fx = 176x + 176^2$

3.82 e0e29be6-463f-477b-a2c1-4960db142fa1 · Hard

The function $f(x) = \frac{1}{9}(x - 7)^2 + 3$ gives a metal ball's height above the ground $f(x)$, in inches, x seconds after it started moving on a track, where $0 \leq x \leq 10$. Which of the following is the best interpretation of the vertex of the graph of $y = f(x)$ in the xy-plane?

A. The metal ball's minimum height was 3 inches above the ground.

B. The metal ball's minimum height was 7 inches above the ground.

C. The metal ball's height was 3 inches above the ground when it started moving.

D. The metal ball's height was 7 inches above the ground when it started moving.

3.83 6e51407b-255f-4375-91cf-b3145e516d19 · Hard

The area of a triangle is equal to x^2 square centimeters. The length of the base of the triangle is $2x + 22$ centimeters, and the height of the triangle is $x - 10$ centimeters. What is the value of x?

Student-produced response: _____

3.84 24e16e9f-c530-4992-98b0-9848ddf76fbb · Hard

The functions f and g are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the maximum value of the function it defines, where $x \geq 0$?

I. $f(x) = 18(1.25)^x + 41$
II. $g(x) = 90.73^x$

A. I only

B. II only

C. I and II

D. Neither I nor II

3.85 a09110a0-a8bc-49cf-ab9f-f198964cf20f · Hard

$$y = 9\left(\frac{a}{7}\right)^{x+c} - b$$

How many times does the graph of the given equation in the xy-plane cross the x-axis, where a, b, and c are positive constants such that $a > 7$ and $b > c$?

A. Zero

B. One

C. Two

D. Three

3.86 6233db14-f861-4237-b90c-a47a7b724601 · Hard

A submersible device is used for ocean research. The function $g(x) = -\frac{1}{55}x + 19x - 35$ gives the depth below the surface of the ocean, in meters, of the submersible device x minutes after collecting a sample, where $x > 0$. How many minutes after collecting the sample did it take for the submersible device to reach the surface of the ocean?

Student-produced response: _____
